



OTTO VON GUERICKE
UNIVERSITÄT
MAGDEBURG

FACULTY OF ECONOMICS
AND MANAGEMENT

OTTO-VON-GUERICKE-UNIVERSITY MAGDEBURG

SEMINAR TERM PAPER

Seminar: Media Economics

Summer Semester 2026

A study on Geopolitical Risk Index and gold returns during war episodes from 1991-2024

Advisor: Jun.-Prof. Shasha Li

Date of Submission: July 10, 2026

Submitted by:

Niranjana Chandran (254396)

Halima Astaoui (260335)

Nandiinii Maithani (259149)

GitHub repository address : https://github.com/Niranjana24-hub/GPRIndex_Analysis

ABSTRACT

Gold is widely considered as a safe -haven investment, yet it's still uncertain whether political risk reliably forecasts gold returns or if the forecasting ability varies between realized events and potential anticipated threats. Geopolitical conflicts create significant uncertainty in financial markets, prompting investors to seek out assets capable of preserving their value during periods of high instability. Measuring geopolitical risk, this study uses the Caldara-Iacoviello Geopolitical Risk Index (GPRI) and its two components: (GPRA) and (GPRT). In terms of techniques, it applies ordinary least squares (OLS) method with Newey-West standard errors to address autocorrelation and heteroscedasticity, Horse Race regression and the Wald test of equality to compare the distinct predictive powers of GPRA and GPRT, as well as GARCH-in-Mean (GARCH-M) and Dynamic Conditional Correlation (DCC) GARCH models, in order to study gold's safe-haven properties and portfolio scope. The study used monthly data from January 1991 to December 2024 capturing three major events: the Gulf War, the Russia-Ukraine War in 2022 and 9/11 attacks. According to the empirical analysis, geopolitical risk helps predict gold returns primarily during periods of high geopolitical tension, while this relationship weakens during periods of calm, which supports the idea that gold serves as a safe-haven asset specifically during armed conflicts. According to the horse race analysis, geopolitical threats have greater predictive power than actual geopolitical events, as investors react to anticipated conflicts even before hostilities begin. Through the DCC-GARCH model, the diversification benefits of gold are confirmed because a persistent negative correlation with equity returns is shown, which becomes even stronger during major geopolitical crises. In conclusion, these findings contribute to the literature on safe-haven assets and the economics of the media, as they demonstrate that geopolitical risk indicators derived from print media provide highly valuable information for predicting the dynamics of the gold market when geopolitical uncertainty is high.

Keywords: Geopolitical Risk Index, gold, safe haven asset, GPRA, GPRT, DCC-GARCH, GARCH-in-Mean (GARCH-M).

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1. INTRODUCTION

Throughout the history of mankind, times of war, armed conflict, and terrorist attacks always bring higher geopolitical and financial instability, which causes market uncertainty to grow and investor confidence to fall (Caldara & Iacoviello, 2022). Because of these unstable conditions, investors are forced to reduce their holdings in risky financial assets so that their money can be moved to safer assets that can preserve value (Baur & Lucey, 2010; Baur & McDermott, 2009). Majority of people believes that gold serves as a reliable protection for wealth because gold possesses international recognition, strong liquidity, and minimal risk of default (Baur & McDermott, 2009; Triki & Maatoug, 2021). Due to its unique ability to hedge against inflation while providing diversification benefits and protection against default risk, gold is considered to be one of the most reliable havens during geopolitical crises (Aizenman & Inoue, 2013).

Geopolitical tensions like the 1991 Gulf War, the September 11, 2001 terrorist attacks, and the 2022 Russia-Ukraine conflict show how gold functions during crises, because rising uncertainty always increases the demand for assets that can preserve their value (Baur & Lucey, 2010; Caldara & Iacoviello, 2022). It has been observed that modern improvements in measuring geopolitical risk now allow researchers to analyze this relationship in a more systematic way. For this purpose, the Caldara–Iacoviello Geopolitical Risk (GPR) Index is used by analysts to evaluate global tensions, which helps them understand how major international events influence the decisions of investors and the price movements in the gold market (Caldara & Iacoviello, 2022). More specifically, the Caldara-Iacoviello Geopolitical Risk Index (GPR) assesses this risk based on media coverage from major international news outlets, making it possible to examine how geopolitical events reported by the media influence investor behavior and financial markets.

Under these circumstances, our study aims to answer two research questions; does the geopolitical risk index (GPRI) derived from newspaper articles can predict gold returns during periods of high geopolitical risk? and does confirmed geopolitical incidents (GPRA) and geopolitical threats (GPRT) exhibit statistically different predictive power for gold returns?

This study therefore analyzes the relationship between geopolitical risk and gold returns between 1991 and December 2024, based on 407 monthly observations. The objectives are to assess whether the GPR Index can be used to predict gold returns during crisis period and whether the GPRA and GPRT exhibits distinct predictive capabilities. The study carries ordinary least squares (OLS) regression, Horse Race regression accompanied by Wald tests, GARCH (1,1)-in-Mean models, and DCC-GARCH models (conditional correlation).

This study is organized as follows: Part 1 introduces the general topic, the rationale for choosing this topic and the research question followed by review of relevant literatures. Part 3 describes the data and the methodology used to conduct the empirical analysis. Part 4 presents and discusses the results. Part 5 concludes the article and offers recommendations for action.

2. LITERATURE REVIEW

This section reviews the key studies that engaged in studying the relationship between geopolitical risk and safe-haven demand.

Gold has widely attracted many scholars due to the tendency of remaining a stable asset even during the times of Market stress where other assets falter. This observation led to the definition of the characteristics of the Safe Haven assets (Baur & Lucey, 2010). Some researchers trace Gold's role as a shock absorber even four decades ago, across different developing countries also which included Brazil, India, Turkey (Ming et al., 2023). With that being said, the evidence was extended to the G7 countries, China and India, proving the fact that Gold provided protection against the dot-com burst, the 2008 financial crisis and the Covid-19 pandemic (Gomes et al., 2023). Similarly, recordings showed that Gold was performing a stabilizing role for developed economies during the 2008 crash (Baur & McDermott, 2009). More recent study by (Manzli & Jeribi, 2024) show that gold even surpassed Bitcoin as a safe-haven asset during the Russia -Ukraine War and the Silicon Valley Bank collapse.

Geopolitical risk is noted to represent a related but distinct channel through which the uncertainty leaks into particularly the gold markets. The picture is further broadened by (Adeosun et al., 2023), by showing that geopolitical risk and policy uncertainty affects return across the precious metals complex, with gold serving as the classic safe haven, even though it is found to have weaker effects than platinum and palladium. The causal ordering is further pushed into finding that policy uncertainty tends precede, rather than follow movements in the price of gold (Amos & Zhang, n.d.). The study by (Engelberg & Parsons, 2011) goes further in the same vein and finds evidence for causal relationship between local news about a particular stock and local trades, which is the process through which media affect market behaviors. This is the basis upon which the current research is extended into the realm of geopolitics.

Media have a causal effect on investor decisions; thus, one can assert that: GPR newspaper coverage creates fear in investors and makes them invest in gold to protect themselves from geopolitical risk. The operationalization of the above fact will be carried out using Geopolitical Risk (GPR) index of (Caldara & Iacoviello, 2022), which is a textual measure of geopolitical risk comprising of two indices; GPRA (geopolitical action) and GPRT (geopolitical threats). According to (Caldara & Iacoviello, 2022), GPR shocks raise the gold price only when the risk is high.

However, after reviewing all the studies, it is found that there is absence of a particular procedure for testing the fact that the impact of GPRA on gold returns is like the impact of GPRT, which is done in this research by conducting the Wald test. In the past, research would study GPRA and GPRT separately in the form of regression analysis, which will not be able to establish the differential predictability formally. Furthermore, in the past, machine learning methods were applied in determining the predictability of GPRA and GPRT. This is the gap in the literature that is to be bridged by this research. The significance of the finding from this research ($p=0.007$) is

that this is the first empirical statistical proof that the event of war and the threat of war are differently predictable in gold returns.

3. DATA

As for this study's dataset, we analyze the period from January 1991 to December 2024, a total of 407 monthly observations which includes the major events such as, the Gulf War (1991), the 9/11 terrorist attacks (2001), and the Russia–Ukraine invasion (2022).

In particular, the GPR index was created by (Caldara & Iacoviello, 2022) through automated computer searches of the online archives of ten major newspapers, including the Chicago Tribune, the Daily Telegraph, the Financial Times, The Globe and Mail, The Guardian, the LA Times, The New York Times, USA Today, The Wall Street Journal, and The Washington Post.

To begin with, a baseline estimate was constructed by calculating the average proportional share for all ten newspapers and normalizing it to a mean of 100 for the 1985–2019 base period. Subsequently, if the index value for a given month equals 120, this suggests that geopolitical risks are covered in newspapers 20 percent more intensively than on average across the whole period under analysis. Similarly, an index value of 513, which was observed in September 2001, implies that the coverage of geopolitical risk was more than five times intense than usual for the majority of newspapers. Nevertheless, in order to ensure methodological transparency, the index can be replicated because it is presented in an open access format downloadable from matteoiacoviello.com on a month-to-month basis since January 1985

3.1. Data sources

Information about the six data series used in this research is provided in Table 1 below, which includes their sources, periods, and additional notes

Table 3.1 Data Sources and Coverage

Variable	Source	Period	Notes
Gold spot price	GitHub: datasets/gold-prices (World Gold Council/ICE)	Jan 1991–Dec 2024	FRED permanently deleted gold data Jan 2022. The GitHub dataset covers the gold price history from 1833 to today.
S&P 500 index	Yahoo Finance (GSPC)	Jan 1991–Dec 2024	The FRED SP500 series starts in 2016, while Yahoo Finance has data from 1991.
VIX	FRED: VIXCLS	Jan 1991–Dec 2024	CBOE Volatility Index traded on the CBOE. Daily data is resampled to monthly average. Available from Jan 1990.
GPR benchmark	matteoiacoviello.com	Jan 1991–Dec 2024	Constructed from 10 newspaper archives. Available from 1985.
GPRA sub-index	matteoiacoviello.com	Jan 1991–Dec 2024	Acts: Categories 6 (war onset), 7 (escalation), 8 (terror acts).

GPRT sub-index	matteoiacoviello.com	Jan 1991– Dec 2024	Threats: Categories 1-5 (war threats, buildups, nuclear, instability).
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3.2. Sub-sample Design

Out of the whole 407-month panel, two sub-samples are built by the analysis; war sub-sample (N = 76), which includes those months where the GPR benchmark is greater than 120 and control group (N = 116), where GPR stays lower than 80 while no episode dummy is active, which represents truly quiet times that have small geopolitical media coverage. Further, a middle-ground of 215 months (GPR between 80 and 120) is removed from both categories.

3.3. Variable Definitions

Table 2 provides a condensed summary of all variables employed in the four estimation methods. Below are their definitions and economic justifications.

Table 3.2 Complete Variable Reference

Variable	Symbol	Type	Construction	Role in Model
Gold Return (lead)	$R_{GOLD,t+1}$	Dependent	$\ln \left(\frac{P_{gold,t+1}}{P_{gold,t}} \right) \times 100$	Next-month gold return (prediction target)
GPR Benchmark	$\Delta \ln (GPR_t)$	Key regressor	$\ln (GPR_t) - \ln (GPR_{t-1})$	Overall geopolitical risk change
GPR Acts	$\Delta \ln (GPRA_t)$	Key regressor	$\ln (GPRA_t) - \ln (GPRA_{t-1})$	Realized conflicts (wars, terror acts)
GPR Threats	$\Delta \ln (GPRT_t)$	Key regressor	$\ln (GPRT_t) - \ln (GPRT_{t-1})$	Anticipated geopolitical tensions
S&P 500 Return	$R_{SP500,t}$	Control	$\ln \left(\frac{SP500_t}{SP500_{t-1}} \right) \times 100$	Equity market control
VIX	VIX_t	Control	Monthly average level	Financial uncertainty control
War sample	WAR_HIGH_t	Sub-sample	$GPR_t > 120$ or crisis period	High-risk regime indicator
Control sample	$CONTROL_t$	Sub-sample	$GPR_t < 80$, no episode	Low-risk regime indicator

4. METHODOLOGY

This paper is structured around four related approaches that together aim to build a detailed and rigorous analytical framework.

4.1. Ordinary Least Squares Regression with Newey-West Standard Errors

Estimation through minimizing squared residuals using Newey-West HAC standard errors for dealing with autocorrelation and heteroskedasticity in the monthly financial data series.

Examination of the direction and magnitude of the relationship between GPR and gold on the full data sample, sub-sample of war period, and control group, within this mathematical framework as follows:

$$R_G^{OLD}_{(t+1)} = \alpha + \beta^1 \cdot \Delta \ln(GPR_t) + \beta^2 \cdot R_{SP500}_t + \beta^3 \cdot VIX_t + \varepsilon_{(t+1)}$$

Using next-month gold returns helps to avoid contemporaneous endogeneity, and this allows the models to capture the predictive content of geopolitical risk.

4.2. Horse Race Regression and Wald Equality Test

To evaluate the independent impact of the two geopolitical risk factors, GPRA and GPRT, the horse race regression is conducted. In this case, the two variables are included in the same regression equation with S&P 500 and VIX being controlled. We argue that in such a way, the differential impact of the two risks on the dependent variable can be isolated. The model takes the following form

$$R_{t+1}^{GOLD} = \alpha + \beta_1 \Delta \ln \ln (GPRA_t) + \beta_2 \Delta \ln \ln (GPRT_t) + \beta_3 R_t^{SP500} + \beta_4 VIX_t + \varepsilon_{t+1}$$

This method isolates one of the components that has predictive power for future gold returns after controlling for their common information. Wald test was conducted by treating the coefficients on GPRA and GPRT as equal constraints to test the null hypothesis:

$$H_0: \beta_1 = \beta_2$$

4.3. GARCH-in-Mean (GARCH-M)

To measure the changing volatility inside gold returns, the GARCH (1,1)-in-Mean model is employed to see if geopolitical risk has an effect on expected returns and volatility of returns. The GARCH (1,1)-in-Mean model makes the standard GARCH framework wider because it permits conditional volatility to go inside the mean equation. Through this way, this model can hold the possible risk-premium channel of volatility while the direct effect of GPR on returns also exists. This specific GARCH (1,1)-in-Mean model is written in this manner:

$$R_t^{GOLD} = \mu + x_0 \Delta \ln \ln (GPR_t) + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

$$z_t \sim \text{Skewed Student} - t(0, 1, \eta, \lambda)$$

$$\alpha + \beta < 1$$

4.4. Dynamic Conditional Correlation GARCH

The Dynamic Conditional Correlation (DCC) GARCH Model was introduced by (Engle, 2002) and serves as a multivariate generalization of the GARCH model. It involves a time varying conditional correlation, ρ_t and, allows one to capture the relationship between asset returns which evolves over time. This is useful in our case, as it can help identify whether the

interplay between gold and stocks is different in the case of war, thus helping to capture the safe haven effect, the model is specified as:

Stage 1 – Univariate GARCH:

$$R_t^{GOLD} = \mu_G + \varepsilon_{G,t}, \quad \sigma_{G,t}^2 = \omega_G + \alpha_G \varepsilon_{G,t-1}^2 + \beta_G \sigma_{G,t-1}^2$$

$$R_t^{SP500} = \mu_S + \varepsilon_{S,t}, \quad \sigma_{S,t}^2 = \omega_S + \alpha_S \varepsilon_{S,t-1}^2 + \beta_S \sigma_{S,t-1}^2$$

Standardized residuals:

$$z_t = \frac{\varepsilon_t}{\sigma_t}$$

Stage 2 – DCC process:

$$Q_t = (1 - a - b) \underline{Q} + a (z_{t-1} z'_{t-1}) + b Q_{t-1}$$

$$\rho_t = \frac{Q_{12,t}}{\sqrt{Q_{11,t} Q_{22,t}}}$$

Stationarity condition:

$$a + b < 1$$

5. DISCUSSION AND FINDINGS

5.1. Ordinary Least Square regression

OLS helped in estimating the magnitude and direction of GPR-gold relationship across the three subsamples (full sample (N = 406), war sub-sample (N = 75), and control group (N = 116) in our present study.

Table 5.1 OLS Regression

Variable	Full Sample	War Sub-sample	Control group
N	406	75	116
$\Delta \ln(\text{GPR}_t)$	0.377	0.976	0.538
NW SE	(0.600)	(0.956)	-1410
p-value	[0.531]	[0.311]	[0.704]
R_SP500_t	0.010	0.210**	0.015
NW SE	(0.041)	(0.081)	(0.068)
VIX_t	0.018	0.015	-0.002
Constant	0.141	0.224	0.237
R ²	0.002	0.083	0.001

Note: *** p<0.01, ** p<0.05, * p<0.10.

The GPR Index has no predictive power over the full sample as the total number of observations is pooled from 76 war months observations and 331 (215 middle-ground and 116 control group obs.) non-war observations. As a result, full sample doesn't represent the relationship, but war sample does, which is as follows.

Unlike full sample results, war sample results show positive difference. The intensified coverage of war by major newspapers triggers the investors to buy gold as insurance against equity market losses and other uncertainties. A one percent increase in the GPR Index (percentage point rise in newspaper relative to previous month) is associated with a 0.976% increase in gold returns which is noteworthy, although this doesn't reach conventional significance ($p < 0.10$). Also, the $R^2 = 0.083$, unlike full sample ($R^2 = 0.002$) and control group ($R^2 = 0.001$) explaining 8.3% of variation in gold returns across the sample period. Caldara and Iacoviello (2022) in their study documented that GPR shocks generate positive gold price responses with effects concentrated during high political geopolitical risk episodes. Also, (Triki & Maatoug, 2021) in their research paper found positive results for GPR index- gold asset using DCC-GARCH analysis, for which, they laid base on OLS estimation.

Another noteworthy result is the control variable R_SP500 , which is found statistically significant ($\beta_2 = 0.210$, $p = 0.011^{**}$), indicating that both gold and equities can move positively together in response to macroeconomic conditions during crisis. The GPR Index coefficient's ($p = 0.311$) failure to reach significance regardless its meaningful magnitude is plausibly a statistical power issue: the reduced war sample, $N=75$ together with the NW corrected standard errors elevates due to the autocorrelation correction

In the control group, gold returns approximately 0.24% per month despite what GPR Index ($\beta_1 = 0.538$, $SE = 1.410$ and $p\text{-value} = 0.703$) does. Gold is purchased by investors for various routine reasons including jewelry demand, industrial demand, portfolio rebalancing etc that operates without geopolitical risk.

5.2. Horse race Regression and Wald test

The horse race is to test whether GPRA (confirmed war acts) or GPRT (anticipated war threats) has higher independent predictive power for gold returns, by allowing them to compete each other, although running separate regressions would not reveal which one is more informative since both are positively correlated with the overall GPR Index. A Wald test of null hypothesis $H_0: \beta(GPRA) = \beta(GPRT)$ formally evaluates whether the two carry statistically distinguishable predictive power.

Table 5.2 Horse Race regression and Wald test

Variable	GPRA Only	GPRT Only	Full Sample	War Sub-sample
$\Delta \ln(GPRA_t)$	0.104	—	-0.081	-1.437*
NW SE	(0.480)	—	(0.683)	(0.816)
p-value	[0.828]	—	[0.906]	[0.083]
$\Delta \ln(GPRT_t)$	—	0.378	0.427	2.942**
NW SE	—	(0.655)	(0.892)	-1137
p-value	—	[0.565]	[0.633]	[0.012]
R_SP500_t	0.009	0.010	0.009	0.244***
Wald test	—	—	$p=0.856$	$p=0.007^{***}$

R^2	0.001	0.002	0.002	0.107
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Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

In the full sample (332 non-war months) investor behavior towards gold is driven by neither GPRA (coefficient- 0.104, $p = 0.828$, NW SE = 0.480) nor GPRT (coefficient - 0.378, $p = 0.565$, NW SE = 0.655) rather the routine gold movements such as jewelry demand, inflation expectations, equity market momentum etc. Although, the coefficient of GPRT is 3.6 times larger than GPRA pointing that on an average, the threat coverage has greater raw predictive power than acts coverage.

When entered together, GPRA turns negative (-0.081 , $p = 0.906$) while GPRT remain positive (0.427 , $p = 0.633$, signing the classic multicollinearity problem, since positive predictive power is being attributed to GPRT and the residual to GPRA by OLS estimator. The negative sign could go deeper economically rather than a statistical artefact. GPRT was already high in several months where threats are elevated (example: 9/11, elevated terrorism threat coverage) and the spike of GPRA in September 2001 (original attack) where the gold prices rose initially but resolved after the initial panic. So, this signals that the confirmed acts can resolve the uncertainty partially by moving it from threat to reality, reducing the Knightian uncertainty. Finally, both GPRT and GPRA are significantly zero predictors of gold as proved by the Wald estimator which yields $p = 0.856$ which is far from rejection of $H_0: \beta(\text{GPRA}) = \beta(\text{GPRT})$. Further, the failure of Wald test can also be attributed because of sample heterogeneity.

However, the war sample got much more to say. A rational investor behaves in such a way that they respond to newspaper threat coverage (GPRT) in advance, pre-positioning in gold. They adjust gold allocation without waiting for war confirmation anticipating gold as a safe-haven channel. The values of GPRT ($\beta = 2.942$, $p = 0.012^{**}$, NW SE = 1.137, $t = 2.588$) are positive and statistically significant defining that a one percent increase in GPRT during a war month is associated with a 2.942 % increase in the following months' gold return. A cleanest illustration is given by Russia-Ukraine episode where gold rose +2.1 % in December 2021 and +4.15 in January 2022 specifically on GPRT, before when the actual invasion began in February 2022.

The GPRA, on the other hand, has values ($\beta = -1.437$, $p = 0.083^*$, NW SE = 0.816, $t = -1.760$) that is negative and marginally significant consistent with “buy the rumor, sell the news”.

Once gold has already bought in anticipation of war (GPRT), the confirmation of later (GPRA) generates a small additional gold return – because threat phase rallied the front-load, not confirmation phase. (Karahana, 2025) in the study found that ambiguity situations generate stronger safe-haven demand than the measurable risk situation confirming GPRT has greater prediction over GPRA. Similarly, (Pástor & Veronesi, 2012) showed empirically that an asset price premium commended by political uncertainty diminishes when the uncertain event is certain or confirmed.

Also, R^2 explains 10.7 % of war month- gold return variation, which is meaningful compared to the R^2 explained in full sample (0.002) and Wald test strongly rejects equality of coefficients: $H_0: \beta(\text{GPRA}) = \beta(\text{GPRT})$ ($p = 0.007$) conforming that both have different predictive power for gold returns during war periods. Overall, despite whether investors buy gold in anticipation of war (GRPT) or as a reaction to war onset (GRPA), the outcome is same: gold

delivers significant positive returns under a geopolitical risk. With the information on GPRT that already encodes markets fear of war, GPRA adds relatives' little information that has not already being priced into gold.

5.3. Generalized Auto Regressive Conditional Heteroscedasticity GARCH (1,1)

A preliminary ARCH LM test on gold returns resulted in a coefficient of 24.99 with $p=0.015$, rejecting the null of no ARCH effects at 5% level. This result confirms that gold return variance is time varying and its clusters during crisis periods. Therefore, GARCH model was necessary as it captures both conditional heteroscedasticity and tests whether GPR affects gold through the volatility channel. (Gupta et al., 2024) used GARCH – MIDAS with GPR to forecast gold return volatility at monthly frequency.

Table 5 .3 Generalized Auto Regressive Conditional Heteroscedasticity GARCH(1,1)

Parameter	Coefficient	Std. Error	p-value	Sig.
PANEL A: Mean Equation				
Constant (α)	0.408	0,171	0,017	**
x_0 ($\Delta \ln \text{GPR}_t$)	0.929	0,651	0,153	
PANEL B: Volatility Equation				
omega (ω)	1,279	0,832	0,124	
alpha[1] (α)	0,138	0,055	0,012	**
beta[1] (β)	0,756	0,102	<0.001	***
$\alpha + \beta$	0,894	—	—	
PANEL C: Distribution				
eta (η)	8,391	3,497	0,016	**
lambda (λ)	0,165	0,071	0,019	**
MODEL FIT				
Log-Likelihood	-1056.64	—	—	
AIC	2127,28	—	—	
N (observations)	407	—	—	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

In the mean equation, the gold generates a positive monthly return of approximately 0.41%, even in the absence of geopolitical risk, ($\alpha = 0.408$, $p = 0.017^{**}$) -gold's resting rate of return. But the GPR coefficient is not significant at the conventional levels ($x_0 = 0.929$, $p = 0.153$) even though economically meaningful in magnitude and direction. A 1% increase in the GPR is associated with a 0.929 % increase in the expected gold return in the same month.

The volatility equation shows 13.8 % of previous month's squared shock to gold returns flows into current month's conditional variance, as predicted by ARCH results ($\alpha [1] = 0.138$, $p = 0.012$, $t = 2.501$). for instance, the attack of September 2001; the coefficient of alpha says that

13.8% of that shock's intensity directly elevates the next month's (October) volatility. Similarly, GARCH (beta [1] = 0.756, $p < 0.001$, $t = 7.438$) signifies that 75.6 % of last month's conditional variance persists into this month.

Total persistence of the volatility, $\alpha + \beta = 0.894 < 1$, confirms covariance stationarity, which means it is mean-reverting with a half -life period of 6.2 months ($\ln(0.5) / \ln(\alpha + \beta) = \ln(0.5) / \ln(0.894) \approx 6.2$ months). Those investors who buys gold when war breaks out is not just benefiting from one month of elevated returns, but half a year.

The distribution parameters indicate substantial leptokurtosis ($\eta = 8.391$ ($p = 0.016^{**}$); extreme monthly returns such as once in Ukraine invasion (Feb 2022: 6.2%) and the other after 9/11 (Sept, 2001: 5.8%) occurred far frequently than a normal distribution would predict reflecting the reality of gold's crisis driven behavior and not just outliers that broke the model. Also, the positive skewness ($\lambda = 0.165$, $p = 0.019$) confirms gold is asymmetrically rewarding as a safe-haven asset.

5.4. DCC – GARCH Dynamic Conditional Correlation GARCH model

A fundamentally different purpose is served by DCC-GARCH analysis compared to the OLS and GARCH-M specifications. While those methods address GPR- gold returns prediction, this model addresses the genuineness of gold as a safe haven asset especially in a portfolio context. It checks the direction of gold during war episodes when equity portfolios are under stress. The confirmation of a negative ρ_t says that gold is a consistent diversifier in equity portfolios.

(Baur & Lucey, 2010) used DCC-GARCH in their study and figured out that during extreme negative stock returns, gold's correlation with stocks turned negative. Similarly, (Triki & Maatoug, 2021) used DCC model and found that higher GPR reduces the gold-equity correlation.

Table 5.4 DCC GARCH

Parameter	Gold Returns	S&P 500 Returns
Panel A: Univariate GARCH(1,1)		
alpha[1] — ARCH coefficient	0.146	0.215
beta[1] — GARCH coefficient	0.738	0.738
alpha + beta	0.884	0.953
Volatility half-life (months)	5.6	14.5
Panel B: DCC Parameters		
alpha_DCC	0.009	
beta_DCC	0.480	
alpha_DCC + beta_DCC	0.489	
Panel C: Time-Varying Conditional Correlation (ρ_t)		
Mean ρ_t	-0.0881	
Std Dev ρ_t	0.0106	
Minimum ρ_t	-0.1466	
Maximum ρ_t	-0.0290	

9/11 episode ρ_t	-0.0951	
Ukraine episode ρ_t	-0.0870	
Panel D: Portfolio Implications		
Mean hedge ratio	-0.0806	
Mean w_{gold}	0.5648	
Maximum w_{gold}	0.8624	
Minimum w_{gold}	0.1503	

Both series are covariance stationary; gold returns ($\alpha[1] = 0.146$ and $\beta[1] = 0.738$, with $\alpha + \beta = 0.884$) and S&P 500 returns ($\alpha[1] = 0.215$ and $\beta[1] = 0.738$, with $\alpha + \beta = 0.953$) but equity volatility represent more persistent (half-life = $(\ln(0.5)/\ln(0.953)) \approx 14.5$ months) compared to gold's 6.2 months). However, equity volatility responds more sharply to new shocks, as the ARCH coefficient for S&P 500 is higher compared to gold returns (0.215 vs 0.146). The DCC estimates indicates small adjustment of conditional correlation where, $\alpha=0.009$ and $\beta=0.480$ respectively indicating that war episodes persist for several months before reverting.

The conditional correlation estimates is negative throughout 1991-2024 (mean $\rho_t = -0.088$, SD = 0.011, range -0.147 to -0.029) with no single month among the 34-year study period has shown a positive conditional correlation between gold and S&P 500 returns confirming that gold is a excellent diversifier.

We can see that the mean value deepens during war episodes with $\rho_t = -0.095$ during the 9/11(Sept-Nov 2001) attack episode and $\rho_t = -0.087$ during the Russia-Ukraine evasion (Feb-April 2022) episode respectively, both below the average of full-sample representing intensification of safe-haven during crisis period. The deepest reading of $\rho_t = -0.147$ reflects the activation of safe-haven feature at its maximum during the peak of global financial crisis.

For portfolio construction, minimum-variance portfolio weight for gold averages at 56.5 % (mean $w_{\text{gold}} = 0.565$) indicating that a variance- minimizing investor should invest 56.5 % of their portfolio in gold. (Triki & Maatoug, 2021) found approximately 57% of optimal portfolio for their 1985-2018 sample, which is close to our finding.

Similarly, during the peak war stress, the model (maximum $w_{\text{gold}} = 0.862$) suggests holding 86.2% of gold and 13.8 % equity and during the tranquil periods (minimum $w_{\text{gold}} = 0.150$), model advices to hold only 15% of gold where the diversification benefits may be smaller, but present. The average hedge ration of -0.081 indicates gold as a short hedge against equities. Economically, an investor holding \$10,000 equity portfolio requires approximately \$810 of gold as a hedge under normal conditions but as the negative covariance between the gold and equities deepens during acute war episodes, the hedge ratio rises.

5.5. Robustness check

A total of six checks were run to confirm that the war sample findings do not depend on a specific threshold- namely, an alternative GPR threshold (below 80), a contemporaneous (more than one month ahead) specification and an exclusion of three different war episodes. The war

period is found to be robust across five of the six checks including RC2 and RC3 specifications confirming that effect is not driven by just one conflict. However, RC1 yields a weaker insignificant sign indicating partial sensitivity of war sample but the reason could be plausibly because of the dilution effect. (Table shown in- **APPENDIX**)

6. CONCLUSION

This study concludes that the Caldara-Iacoviello GPR Index predicts gold returns favorably during war-level crisis and remains insignificant during the tranquil period, confirming gold as a war-specific safe-haven asset. The Wald test formally proves that GPRT and GPRA statistically carries different predictive power with GPRT dominating through the expected safe-haven channel. Also, golds volatility half -life shows that the elevated risk premia persist after the shock has happened for a period of six months. Finally, DCC GARCH shows that gold equity correlation is always negative across the study period which confirms gold permanently diversifies equities and intensifies during the war.

However, the study is subjected to small limitations including a small war sample (N=76) and a US market focused data. The study recommends future research extending the analysis to a cross-country panel. Further, from the policy perspective, model suggests that investors should increase their investment in gold by about 70-86% out of their portfolio during a war-level crisis period, as gold is proved as a safe-haven channel.

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8. APPENDIX

Table. AI Transparency

Tool	Member	Purpose	Verification
Claude	Niranjana Chandran	Python script development Data collection automation Literature review suggestions Table design Conceptual clarification Read.me file generation	All outputs verified by running manually- sample sizes, results etc in Python3 Read.me file verified in gitHub
Claude ai ChatGPT	Halima Astaoui	Data preparation econometric explanations variable definition writing	All outputs were verified against the cited academic literature and revised before inclusion in the paper.
Claude Zotero	Nandiinii Maithani	Literature review suggestions Formatting Citation clarifications	All outputs were cross-checked against original sources and verified using Zotero citation records before inclusion.

Table . Robustness

Check	Specification	War sample			Control Group			Robust?
		N	$\beta(\text{GPR})$	p	N	$\beta(\text{GPR})$	p	
Baseline	GPR>120, control GPR<80 dep = R_GOLD_(t+1)	74	0.976	0.311	116	0.538	0.704	—
RC1	Threshold GPR>103 (top tercile war) dep = R_GOLD_(t+1)	135	-0.226	0.828	136	0.413	0.766	Partial
RC2	Contemporaneous return dep = R_GOLD_(t)	75	1406	0.113	116	0.506	0.641	Robust
RC3a	Exclude Gulf War (Jan– Mar 1991)	72	0.792	0.455	116	0.538	0.704	Robust
RC3b	Exclude 9/11 + Afghan (Sep–Nov 2001)	71	1331	0.359	116	0.538	0.704	Robust
RC3c	Exclude Russia-Ukraine (Feb–Apr 2022)	71	0.631	0.469	116	0.538	0.704	Robust